

datum

firmware guide

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Overview

datum uses ZMK firmware, built automatically via GitHub Actions whenever you push changes to config/. No local toolchain needed — the .uf2 files are downloadable directly from the Actions tab.

Getting the firmware

1. Go to github.com/hermatician/datum
2. Click the Actions tab
3. Click the latest successful build run
4. Scroll to the bottom and download the firmware artifact
5. Unzip — you will find `cradio_left-nice_nano-zmk.uf2` and `cradio_right-nice_nano-zmk.uf2`

Flashing

1. Connect the Nice!nano via USB-C
2. Double-tap the reset button — a USB drive called NICENANO appears
3. Copy the .uf2 file onto the drive
4. The controller reboots automatically
5. Flash left half first with `cradio_left-nice_nano-zmk.uf2`, then right half

Modifying the keymap

The keymap lives at `config/cradio.keymap` in the datum repo. Edit it and push — GitHub Actions will build new firmware automatically.

Base layout — Gallium (columnar stagger variant)

datum uses the Gallium layout, optimised specifically for columnar stagger keyboards.

```
b l d c v | j y o u ,  
n r t s g | p h a e i  
x q m w z | k f ' ; .  
NAV SPC | BSPC SYM
```

Home row left: `n r t s` — the most common consonants, all on strong fingers.

Home row right: `p h a e i` — high frequency letters, all on strong fingers.

Learning Gallium

Gallium is a significant departure from QWERTY. Budget 2–4 weeks to reach comfortable typing speed. Use Monkeytype for daily practice.

1. Learn home row first: `n r t s / p h a e i`
2. Add top row: `b l d c v / j y o u ,`
3. Add bottom row: `x q m w z / k f ' ; .`

4. Only then add layers and combos

Layer design

Base layer

Left thumb: tap = Space / hold = NAV Right thumb: tap = Backspace / hold = SYM

```
b l d c v | j y o u ,
n r t s g | p h a e i
x q m w z | k f ' ; .
      NAV SPC | BSPC SYM
```

Layer 1 — Navigation + Numbers

```
1 2 3 4 5 | 6 7 8 9 0
TAB HOME UP END PGUP | BSPC LEFT DOWN RIGHT DEL
ESC -- -- -- PGDN | -- -- -- -- RET
      [held] SPC | BSPC FN
```

Layer 2 — Symbols

```
! @ # $ % | ^ & * ( )
` - = [ { | } ] + _ ;
~ | \ < > | :: " ' : ?
      NAV SPC | BSPC [held]
```

:: as a single key — typed constantly in Rust.

Layer 3 — Function + System

```
F1 F2 F3 F4 F5 | F6 F7 F8 F9 F10
F11 F12 -- -- CAPS| -- MUTE VOLD VOLU --
BOOT -- -- -- -- | -- -- -- -- --
      [held] SPC | BSPC [held]
```

BOOT = ZMK bootloader mode. Both thumbs held activates this layer.

Home row mods

Modifiers live on the home row as tap-hold keys, keeping your fingers on the home row at all times.

```
Left:  n=GUI  r=ALT  t=SHIFT  s=CTRL
Right: p=GUI  h=ALT  a=SHIFT  e=CTRL

tapping-term-ms: 200
flavor: balanced
require-prior-idle-ms: 150
```

Combos

```
J + K simultaneously -> ::
M + , simultaneously -> ->
, + . simultaneously -> =>
```

Build details

Property	Value
Shield	cradio (Ferris Sweep / Cradio)
Board	nice_nano//zmk (Nice!nano V2, Zephyr 4.1+)

ZMK revision	main
Build trigger	push to config/ or workflow_dispatch

Resources

- ZMK docs: zmk.dev/docs
- Gallium layout: github.com/GalileoBlues/Gallium
- Monkeytype: monkeytype.com
- datum repo: github.com/hermatician/datum